



The business problem comes first

How can a restaurant improve lunch service when the sales table is full of numbers but the decision is unclear?

THE PROBLEM

Raw rows do not answer the decision.

- Which item sells most?
- Which item earns the most gross profit?
- What service problem appears in customer notes?

THE RISK

A quick AI answer can be wrong.

- Rows can be read incorrectly.
- Totals can be calculated incorrectly.
- A confident summary is not an audit.

THE GOAL

Turn data into a checked decision.

- Organize the table.
- Verify the calculations.
- Use evidence to choose one action.

What "making sense of data" means

AI can help describe patterns. A spreadsheet or calculator must verify the arithmetic.

WHAT

Read columns before drawing conclusions.

- Units sold shows demand.
- Price and cost show gross profit.
- Customer notes explain context.

WHY

A single number is not enough.

- High sales can still mean low profit.
- A comment can reveal a service problem.
- A total needs a checkable formula.

WITHOUT IT

The decision can be misleading.

- Wrong totals lead to wrong priorities.
- AI may fill gaps with guesses.
- The team cannot explain the recommendation.

Use a five-step data audit

Do the calculation in a spreadsheet first. Then ask AI to explain the verified results.

1. Inspect

Read each column and unit.

2. Calculate

Revenue and gross profit.

3. Verify

Check totals in a spreadsheet.

4. Explain

Ask AI for patterns.

5. Act

Choose one measurable action.

Bad prompt: ask for a conclusion without data rules

The prompt does not tell AI which numbers matter or how to treat arithmetic.

PROMPT TO AVOID

Here is our restaurant data.
What should we improve?

WHY IT FAILS

- No columns or decision criterion are named.
- The answer can become generic advice.
- There is no request to verify totals.

Bad prompt: ask AI to guarantee the math

A language-model answer is not a substitute for a calculation check.

PROMPT TO AVOID

Add all sales rows, tell us the exact profit, and guarantee that your total is correct.

WHY IT FAILS

- It asks for certainty without an audit.
- It gives no formula or cross-check.
- A polished number can still be wrong.

Good prompt: inspect the data before analysis

Use this prompt before making any recommendation.

COPY AND ADAPT

You are a restaurant operations analyst.
Use only the Session 2.1 dataset.

First, create three lists:

1. Columns and what each measures
2. Facts visible in the rows
3. Information missing for a full decision

Do not calculate or invent totals.
End with three questions for the owner.

Session 2.1 restaurant sales data

This fictional instructional CSV is complete enough for the exercises. Use a spreadsheet to verify each calculation.

SALES

What sold?

- Burger: 58 units.
- Wrap: 32 units.
- Salad: 18 units.
- Soup: 12 units.

VERIFIED TOTALS

What did the spreadsheet calculate?

- Revenue: \$1,016.00.
- Gross profit: \$633.60.
- Burger gross profit: \$313.20.

CUSTOMER NOTES

What needs attention?

- Two notes mention a lunch queue.
- One item sold out at 1 PM.
- Pickup speed is a possible issue.

Good prompt: explain verified results

AI explains patterns after the spreadsheet supplies the checked totals.

COPY AND ADAPT

You are a restaurant operations analyst.

Use the Session 2.1 dataset and these verified spreadsheet totals:

- Revenue: \$1,016.00
- Gross profit: \$633.60
- Burger gross profit: \$313.20

Write three bullets:

1. One demand pattern
2. One profit pattern
3. One service issue in the notes

Recommend one two-week action.

Name one metric to track.

Do not recalculate the totals.

Exercise 1: Verify, then explain

Students work in pairs with `data/session\_2\_1\_restaurant\_sales.csv`.

STEP 1

Calculate

- Revenue = units sold x unit price.
- Gross profit = units sold x (price - food cost).

STEP 2

Verify

- Check the total revenue.
- Check the total gross profit.
- Compare with Slide 8.

STEP 3

Explain

- Run Good Example 2.
- Mark every claim as fact or interpretation.

Exercise 2: Choose one small improvement

The data show a long lunch queue and strong burger demand. Choose one action to test for two weeks.

PROBLEM

State one observation

- Lunch queue notes appear twice.
- Burger demand is highest.

ACTION

Choose one response

- Prepare burger ingredients before noon.
- Add one pickup sign.
- Assign one staff member to lunch pickup.

MEASURE

Check whether it worked

- Track average lunch wait time.
- Target: reduce it by one minute.
- State the result after two weeks.

A reliable data-to-decision checklist

Use AI for explanation and drafting. Use a spreadsheet for arithmetic and verification.

1

Inspect

- Know what every column measures.

2

Verify

- Use formulas and check totals.

3

Explain

- Ask AI for patterns in verified results.

4

Act

- Choose one measurable improvement.